

Lab Assignment Questions

1. Hidden Object Recovery
 - a. The object in the image is barely visible (in 1-a and 1-b). Why?
 - b. Explain how will you do and apply:
 - i. Recover the object clearly
 - ii. Make it suitable for edge detection
2. You are given 3 images:
 - a. Identify the noise in these 3 images (2-a, 2-b and 2-c)
 - i. Identify noise type
 - ii. Apply optimal filtering
 - iii. Justify your technique
3. Object Extraction with Imperfect Mask

Task:
Apply techniques to extract the object? Explain How you did?
(in image 3-a – Person, In image 3-b suitcase)
4. The image suffers from:
 - Uneven illumination (bright + dark regions)
 - Loss of details in shadow areas
 - The images are 4-a and 4-b
 - a. You are Asked to
 - i. Correct the illumination
 - ii. Enhance hidden details in dark regions
 - iii. Make the image suitable for further processing (edge detection / segmentation)
 - iv. Justify your approach?
5. Two images are given named 5-a, 5-b and 5-c.
 - a. Explain why this image is degraded?
 - b. Restore the image as much as possible?
 - c. Enhance edges and fine details?
 - d. Make text or object readable?
6. The detected edges of an object are Broken / discontinuous, and Contain gaps and noise (Image 6-a)
 - a. Reconstruct continuous object boundaries
 - b. Remove false edges (noise)
 - c. Prepare the image for object detection or contour extraction
 - d. Explain the technique